

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 1**

**COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN**  
**COURSE CODE: CSA1102**

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### **COURSE OUTCOME COVERED**

**CO2:** Design UML diagrams to represent system requirements and architecture.  
(BL3)

*Assessment Tool Weightage: 40%*

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**QUESTIONS: Any 4**

**Total Marks:40**

Annexure A

### **ANALYTICAL PROBLEM SOLVING**

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#### ***Code of Conduct:***

*I B. Harshitha (192420006) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

#### ***Signature of the student:***

B.Harshitha

## ANALYTICAL PROBLEM SOLVING

### 1. Use Case Diagram Problem

A **Hospital Management System** allows patients to book appointments, doctors to update medical records, and administrators to manage hospital resources. The requirements are not clearly documented.

**Tasks:**

- a) Identify all actors involved in the system.
- b) List the primary use cases.
- c) Define relationships (include, extend, generalization).
- d) Construct a complete Use Case Diagram representing system functionality.

**(10 Marks)**

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### 2. Class Diagram Problem

A **Library Management System** contains entities such as Book, Member, Librarian, and IssueRecord. The relationships among entities are not properly defined.

**Tasks:**

- a) Identify classes with attributes and methods.
- b) Define relationships (association, aggregation, inheritance).
- c) Specify multiplicity between classes.
- d) Design a detailed Class Diagram for the system.

**(10 Marks)**

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### 3. Sequence Diagram Problem

In an **Online Food Ordering System**, a customer selects food items, places an order, makes payment, and receives order confirmation. The interaction flow is unclear.

**Tasks:**

- a) Identify objects involved in the process.
- b) Define message flow between objects.
- c) Represent activation and lifelines.
- d) Construct a Sequence Diagram showing complete interaction flow.

**(10 Marks)**

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#### 4. Activity Diagram Problem

A **Student Admission System** includes application submission, document verification, fee payment, and enrollment confirmation. The workflow lacks proper decision-making steps.

**Tasks:**

- a) Identify all activities in the workflow.
- b) Add decision nodes for approval/rejection.
- c) Include parallel processing where applicable.
- d) Design an optimized Activity Diagram.

**(10 Marks)**

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#### 5. State Diagram Problem

A **Vending Machine System** operates through states such as Idle, Product Selected, Payment Received, Product Dispensed, and Transaction Completed.

**Tasks:**

- a) Identify all possible states.
- b) Define events triggering transitions.
- c) Include initial and final states.
- d) Construct a State Transition Diagram representing system behavior.

**(10 Marks)**

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#### 6. Component Diagram Problem

An **Online Learning Platform** consists of modules such as User Management, Course Management, Assessment Module, Content Delivery Module, and Notification Service.

**Tasks:**

- a) Identify major system components.
- b) Define interfaces between components.
- c) Show dependencies among modules.
- d) Construct a Component Diagram representing system architecture.

**(10 Marks)**

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## 7. Deployment Diagram Problem

A **Smart Traffic Monitoring System** uses roadside sensors to collect vehicle data. The data is transmitted to a central cloud server where traffic authorities monitor congestion through a web dashboard. Alerts are generated during traffic violations and accidents.

### Tasks:

- Identify all hardware nodes involved in the system.
- Specify software components deployed on each node.
- Define communication links between nodes.
- Construct a complete Deployment Diagram representing the system architecture.

(10 Marks)

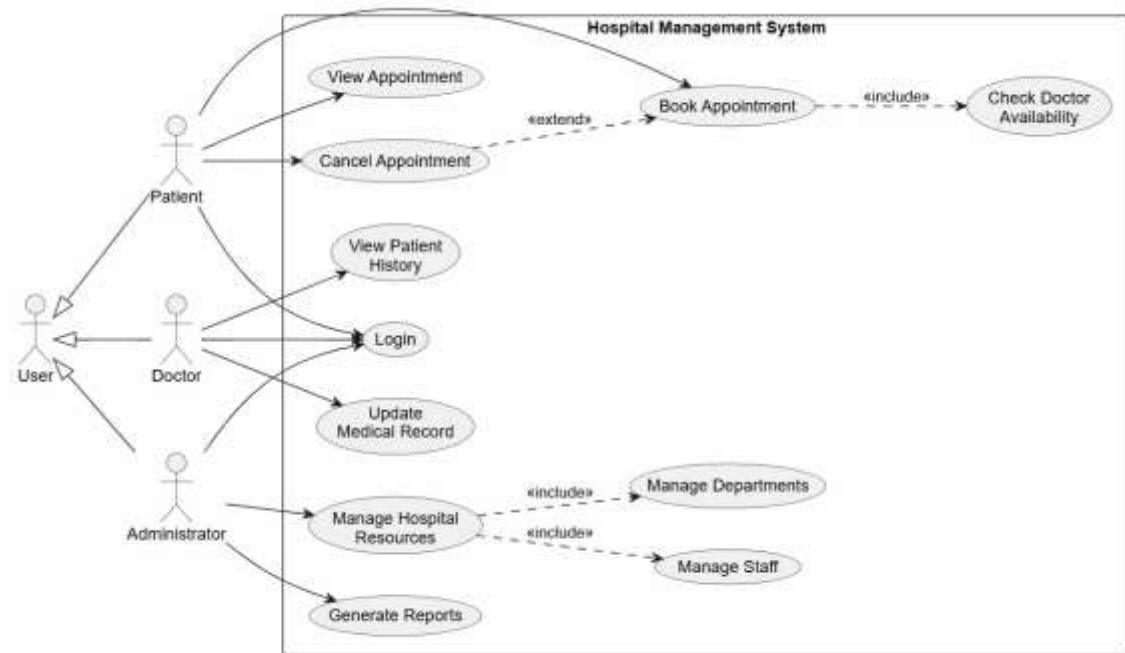
### RUBRICS FOR CALCULATION

#### ANALYTICAL PROBLEM SOLVING

| Criteria                  | Weightage | Level 4 (Exemplary)                                                               | Level 3 (Proficient)                      | Level 2 (Developing)                       | Level 1 (Beginning)                           |
|---------------------------|-----------|-----------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|-----------------------------------------------|
| Problem Understanding     | 20%       | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps       | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection          | 20%       | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors  | Inappropriate or partially correct method  | Unable to select method                       |
| Solution Process          | 25%       | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes  | Disorganized steps; multiple errors        | No clear process                              |
| Accuracy of Calculation   | 20%       | All calculations accurate; correct final answer                                   | Minor calculation errors                  | Frequent errors; incorrect final answer    | No valid calculations                         |
| Interpretation of Results | 15%       | Clearly interprets results with units and engineering relevance                   | Basic interpretation with minor omissions | Limited or unclear interpretation          | No interpretation                             |

## 1. Use Case Diagram Problem

A **Hospital Management System** allows patients to book appointments, doctors to update medical records, and administrators to manage hospital resources. The requirements are not clearly documented.



### a) Actors Involved

- Patient
- Doctor
- Administrator

### b) Primary Use Cases

- Login
- Book Appointment
- View Appointment
- Cancel Appointment
- Check Doctor Availability
- Update Medical Record
- View Patient History
- Manage Hospital Resources
- Manage Staff
- Manage Departments
- Generate Reports

### c) Relationships

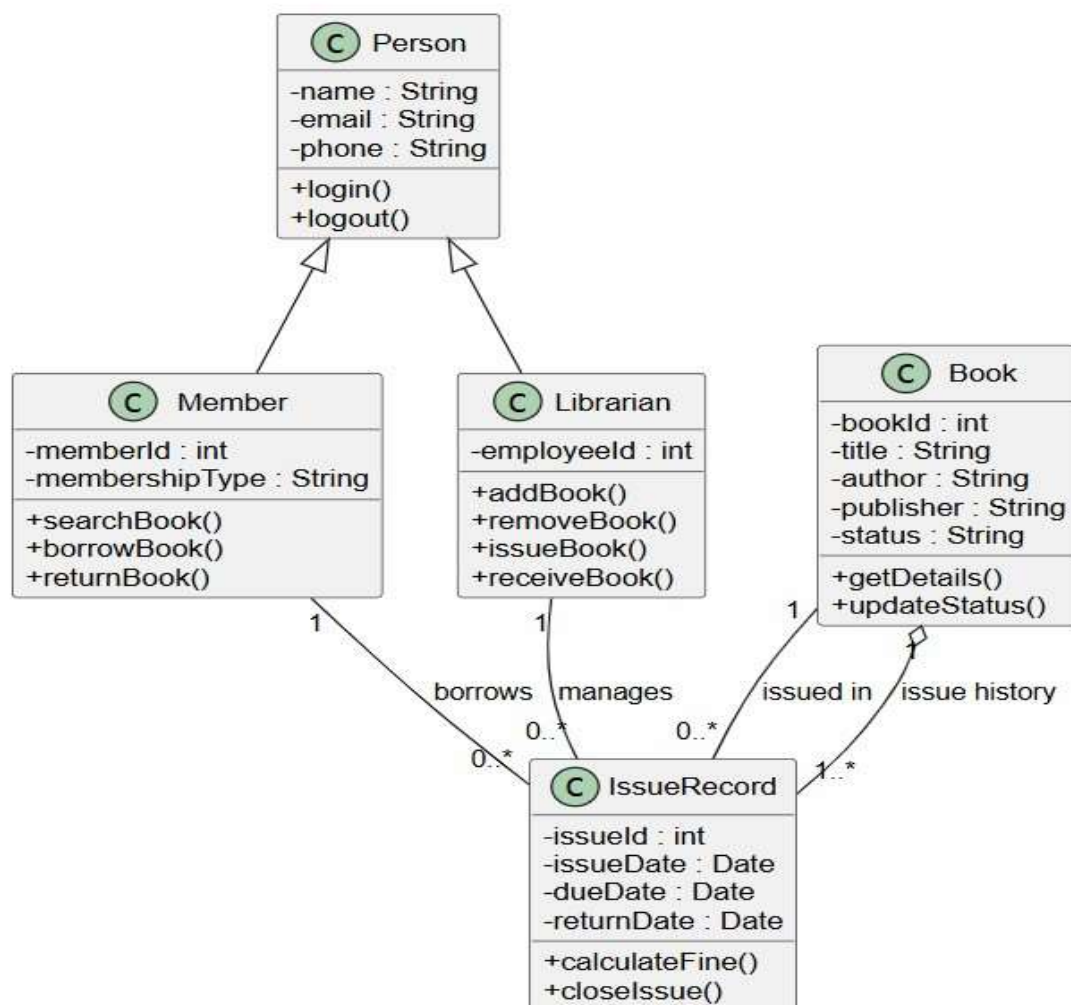
- Association: Actors interact with their respective use cases.
- <<include>>: Book Appointment includes Check Doctor Availability; Manage Hospital Resources includes Manage Staff and Manage Departments.
- <<extend>>: Cancel Appointment extends Book Appointment.
- Generalization: Patient, Doctor, and Administrator inherit from the User actor.

#### d) Conclusion

The Use Case Diagram represents the functional requirements of the Hospital Management System by illustrating the interactions between actors and system functionalities using association, include, extend, and generalization relationships.

## 2. Class Diagram Problem

A **Library Management System** contains entities such as Book, Member, Librarian, and IssueRecord. The relationships among entities are not properly defined.



#### a) Classes with Attributes and Methods

##### Person

- Attributes: name, email, phone
- Methods: login(), logout()

##### Member

- Attributes: memberId, membershipType
- Methods: searchBook(), borrowBook(), returnBook()

##### Librarian

- Attributes: employeeId
- Methods: addBook(), removeBook(), issueBook(), receiveBook()

#### Book

- Attributes: bookId, title, author, publisher, status
- Methods: getDetails(), updateStatus()

#### IssueRecord

- Attributes: issueId, issueDate, dueDate, returnDate
- Methods: calculateFine(), closeIssue()

### **b) Relationships**

- Inheritance: Member and Librarian inherit from the Person class.
- Association: Member borrows books through IssueRecord, and Librarian manages IssueRecord.
- Aggregation: A Book maintains one or more IssueRecords representing its borrowing history.

### **c) Multiplicity**

- One Member can have zero or many (0..\*) IssueRecords.
- One Book can appear in zero or many (0..\*) IssueRecords.
- One Librarian manages zero or many (0..\*) IssueRecords.
- One IssueRecord is associated with exactly one Member, one Book, and one Librarian.

### **d) Conclusion**

The Class Diagram models the static structure of the Library Management System by defining classes, attributes, methods, relationships, inheritance, aggregation, and multiplicity. It clearly represents how books are issued and managed within the library system.

## **3. Sequence Diagram Problem**

In an **Online Food Ordering System**, a customer selects food items, places an order, makes payment, and receives order confirmation. The interaction flow is unclear.

### **a) Objects Involved**

- Customer
- Food Ordering System
- Restaurant
- Payment Gateway
- Notification Service

### **b) Message Flow**

1. The customer selects food items.
2. The system displays the selected items in the cart.
3. The customer places the order.
4. The system sends the order details to the restaurant.
5. The restaurant accepts the order.

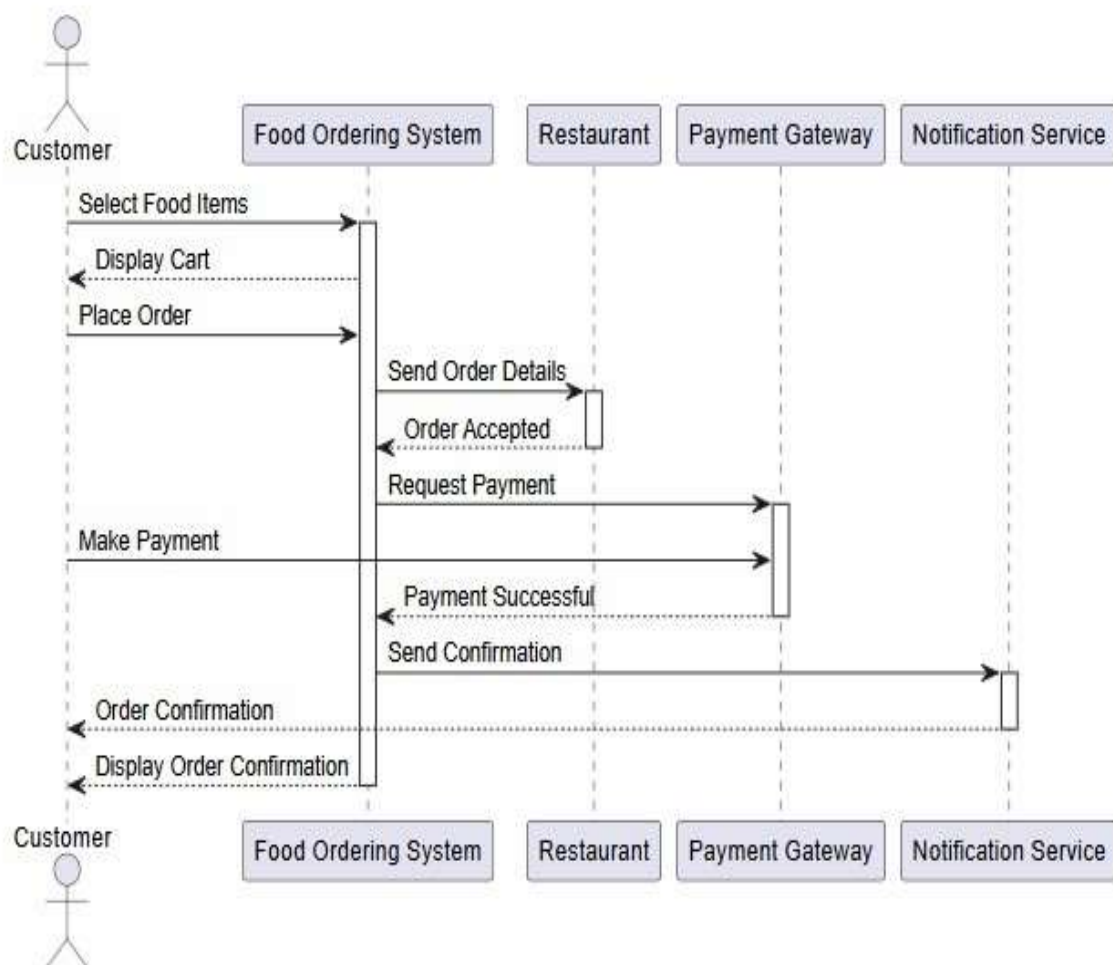
6. The system requests payment through the payment gateway.
7. The customer completes the payment.
8. The payment gateway confirms successful payment.
9. The system sends the order confirmation through the notification service.
10. The customer receives the order confirmation.

### c) Activation and Lifelines

- **Lifelines** represent the existence of each object during the interaction.
- **Activation bars** indicate the period during which an object performs an operation.
- Messages are exchanged sequentially from the customer to the system, restaurant, payment gateway, and notification service.

### d) Conclusion

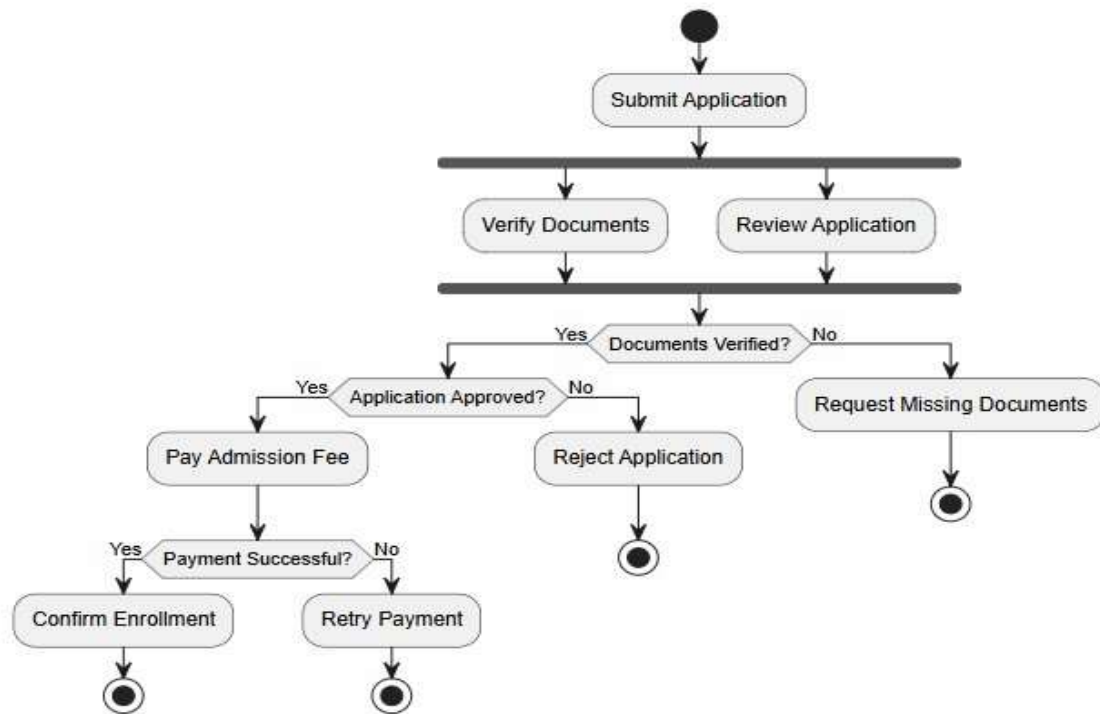
The Sequence Diagram illustrates the complete interaction flow of the Online Food Ordering System. It clearly represents the sequence of messages, object lifelines, and activation periods from food selection to successful order confirmation.





#### 4. Activity Diagram Problem

A **Student Admission System** includes application submission, document verification, fee payment, and enrollment confirmation. The workflow lacks proper decision-making steps.



##### a) Activities in the Workflow

- Submit Application
- Verify Documents
- Review Application
- Pay Admission Fee
- Confirm Enrollment
- Request Missing Documents
- Reject Application
- Retry Payment

##### b) Decision Nodes

- Documents Verified?
  - Yes → Continue to application approval.
  - No → Request missing documents.
- Application Approved?
  - Yes → Proceed to fee payment.
  - No → Reject the application.
- Payment Successful?
  - Yes → Confirm enrollment.
  - No → Retry payment.

**c) Parallel Processing**

The diagram uses parallel processing (Fork and Join) where:

- Document Verification and Application Review are performed simultaneously after the application is submitted.
- Once both activities are completed, the workflow continues to the decision-making stage.

This reduces processing time and improves efficiency.

**d) Conclusion**

The Activity Diagram illustrates the complete admission workflow of the Student Admission System. It includes activities, decision nodes for approval and rejection, parallel processing for document verification and application review, and an optimized flow leading to successful enrollment.